



EskwelaOne: Educational Management Information System for Sindalan National High School

Malana, Laurence Andre' S. • Calma, Xian Meg S. • Turla, Cristopher Ian S.

ABSTRACT

ESKWELAONE IS A LOCALIZED WEB BASED EDUCATION MANAGEMENT INFORMATION SYSTEM (EMIS) DESIGNED TO MODERNIZE AND STREAMLINE STUDENT INFORMATION AND PERFORMANCE MANAGEMENT IN SINDALAN NATIONAL HIGH SCHOOL. BY REPLACING TRADITIONAL MANUAL PROCESSES, THE SYSTEM ENABLES AUTHORIZED SCHOOL PERSONNEL TO SECURELY RECORD, ACCESS, AND UPDATE STUDENT DATA INCLUDING GRADES, ATTENDANCE, BEHAVIOR, AND CLINIC VISITS THROUGH AN INTEGRATED DIGITAL PLATFORM.

UNLIKE PAPER BASED WORKFLOWS THAT REQUIRE REPETITIVE ENCODING AND MANUAL REPORT PREPARATION, ESKWELAONE AUTOMATES KEY ADMINISTRATIVE TASKS, IMPROVES RECORD ACCURACY, AND PROVIDES REAL TIME DATA UPDATES FOR MORE EFFICIENT DECISION MAKING. THE STUDY EMPLOYED A DESCRIPTIVE DEVELOPMENTAL APPROACH SUPPORTED BY A QUANTITATIVE USER EVALUATION USING VALIDATED TAM BASED SURVEY INSTRUMENTS TO ASSESS SYSTEM USEFULNESS, USER ATTITUDE, AND ADOPTION INTENT. FINDINGS REVEALED VERY HIGH RATINGS ACROSS ALL EVALUATION METRICS, CONFIRMING THAT ESKWELAONE SIGNIFICANTLY IMPROVES ADMINISTRATIVE EFFICIENCY, ACCURACY, AND TRANSPARENCY IN SCHOOL OPERATIONS. FUTURE ENHANCEMENTS INCLUDE EXPANDING SYSTEM MODULES, IMPLEMENTING CLOUD BASED DEPLOYMENT, AND INTEGRATING USER PORTALS FOR STUDENTS AND PARENTS TO BROADEN SYSTEM ACCESSIBILITY.

KEYWORDS: ESKWELAONE SYSTEM, SCHOOL OPERATIONS AUTOMATIONS, STUDENT INFORMATION MANAGEMENT, ACADEMIC RECORDS, DISCIPLINE TRACKING, HEALTH PROFILING, RECORD DIGITIZING, REPORT GENERATION

OBJECTIVES



TO REPLACE MANUAL RECORD- KEEPING WITH A CENTRALIZED DIGITAL DATABASE.

TO PROVIDE REAL-TIME DATA ACCESS FOR TIMELY DECISION- MAKING.

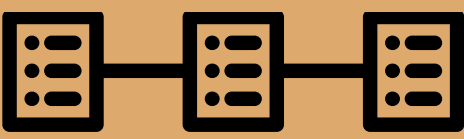
TO REDUCE WORKLOAD THROUGH AUTOMATED REPORT GENERATION.

TO DESIGN A SCALABLE SYSTEM FOR GROWING SCHOOL DEMANDS.

TO IMPROVE OVERALL ADMINISTRATIVE EFFICIENCY AND ACCURACY.

METHODOLOGY (The SDLC Model)

Phase 1: Requirement Analysis: Researchers consulted with the registrar, counselor, and clinic staff to identify challenges in manual workflows



Phase 2: System Design: Followed a user-centered approach and an Input-Process-Output (IPO) model to align with school needs.

Phase 3: Prototype Development: Built core modules for Student Management, Grades, Attendance, and Clinic Records using Django and PostgreSQL.



Phase 4: Quantitative Evaluation: Conducted an evaluation with 10 key personnel using validated TAM instruments to assess system acceptance.

Phase 5: Implementation & Deployment: Successfully deployed the system in a local environment for administrative use, providing a scalable digital infrastructure for Sindalan National High School.

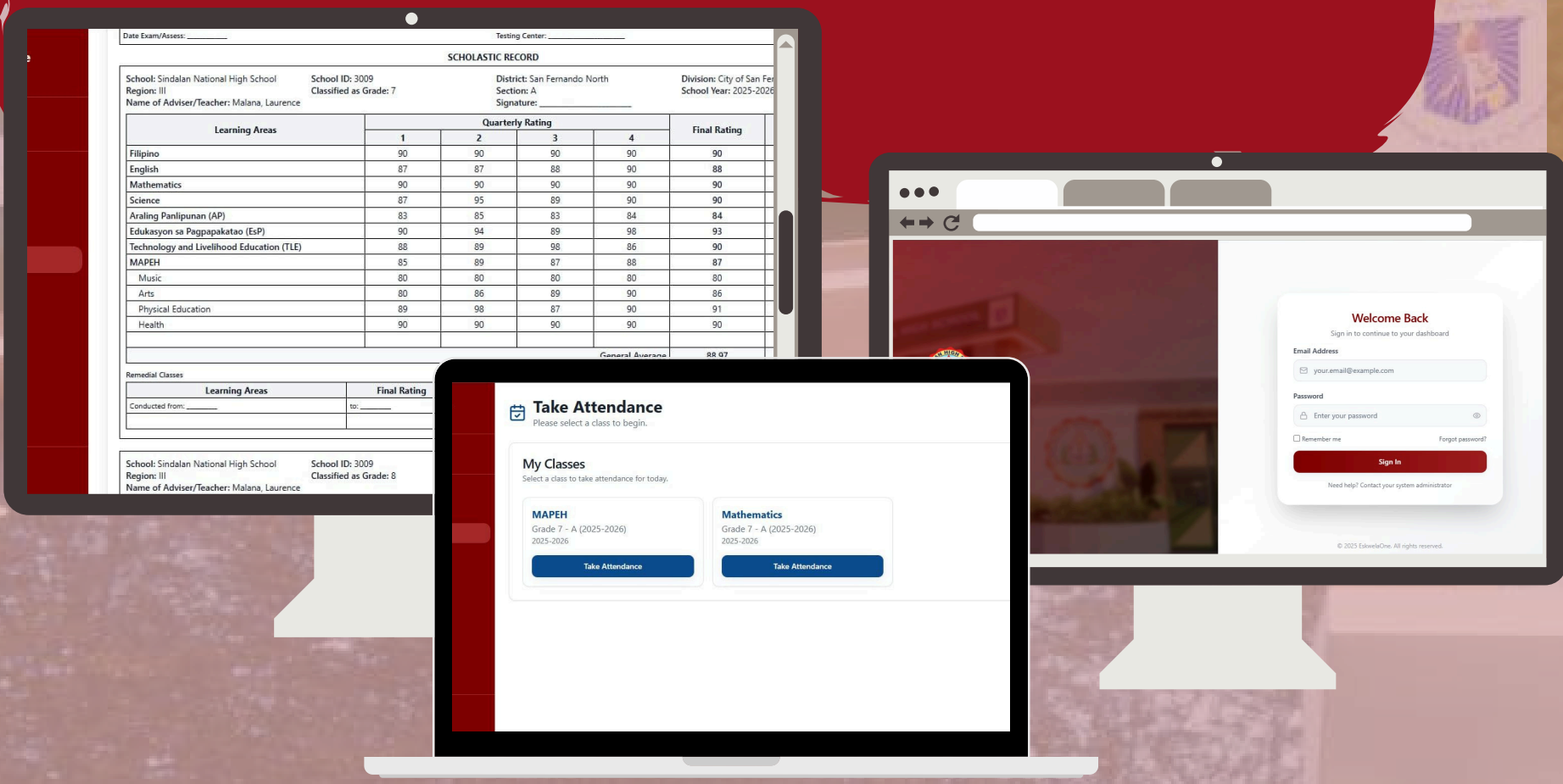


Results

- ✓ EskwelaOne achieved Very High Acceptance (Overall TAM Mean: 3.96)
- ✓ Exceptional Perceived Usefulness (3.99) in automating school records.
- ✓ Personnel showed strong willingness to adopt the system into daily workflows.
- ✓ Features effectively streamlined student tracking, grades, and clinic reporting.
- ✓ System demonstrated technical viability for full-scale school implementation.

Conclusion

- Successfully addresses the systemic inefficiencies and inaccuracies of manual record-keeping at Sindalan National High School.
- Validated as a technically viable and user-friendly EMIS specifically tailored to meet the school's unique administrative needs.
- High user acceptance scores (3.96 Mean) confirm that stakeholders are ready and highly willing to adopt the system for daily operations.
- Empowers decision-makers by providing real-time, accurate, and transparent data for student performance and school management.
- Transforms traditional paper-based workflows into a streamlined, digital, and data-driven administrative environment.



REFERENCES

- Abdul-Hamid, H., Mintz, S., Saraogi, N., & Doran, E. (2017). Data-driven decision making in education: The role of education management information systems (EMIS). World Bank Group.
- Ango, U., Maina, J., & Harbau, M. (2024). Impact of Education Management Information System (EMIS) on planning and decision-making in Nigerian universities. Nigerian Journal of Educational Administration and Planning, 24(1), 45–58.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly, 13(3), 319–340. <https://doi.org/10.2307/249008>
- Department of Education. (2023). Basic Education Development Plan 2030. <https://www.deped.gov.ph/wp-content/uploads/2023/01/BEDP-2030.pdf>
- Venkatesh, V., & Davis, F. D. (2000). A theoretical extension of the technology acceptance model: Four longitudinal field studies. Management Science, 46(2), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>